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New Methods to Improve the Performance of Open and Closed Loop Fiber-Optic Gyros

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Abstract—The paper describes the concept of an open loop fiber-optic gyro (FOG) with digital output and improved scale factor stability. This stability is reached by stabilizing the amplitude of rotation signal and biasing modulation depth. FOGs with closed first and second feedback loops are described, and method of dead zone elimination are outlined. A method of reducing the setup time for gyros with open first feedback loop and with two closed feedback loops by correcting their output is presented. Compensation of parasitic effects in phase modulators of integrated optic chip is proposed as a further perspective of improving FOG accuracy.

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INTRODUCTION

Fiber-optic gyros are extensively used in high accuracy navigation and stabilization problems. This type of gyros offers a number of advantages over mechanical and ring laser gyros in terms of cost, size, power consumption, etc. Interferometric FOGs with open [1] and closed feedback loops [2, 3] are known.

The principal problem of open loop FOG is the instability of scale factor [1, 4] and biasing modulation (BM) depth, which is used to improve rotation sensitivity. These problems can be solved with sinusoidal phase BM, which uses several harmonics at BM frequency in FOG output for data processing. In this case, electronic processing schemes using analog components are usually applied, which are rather complicated but fail to provide the required accuracy.

A FOG with a closed loop, which linearizes its output is described in [2, 3]. Sawtooth voltage (STV) is applied to the phase modulator to compensate for the Sagnac phase difference. To stabilize the scale factor, voltage peak-to-peak amplitude corresponding to phase difference 2π rad between the waves of ring interferometer is maintained [2, 3].

Usually closed loop FOG utilizes rectangular BM [2, 3]. In this case the disadvantages are the dead zone and large setup time. In existing data processing schemes, the dead zone can be caused by electric cross-talk [3], parasitic Michelson interferometer [5], and instability of BM parameters (see Section 2). Scale factor instability at low rates is due to the fact that STV has a large period, so the data used to control BM and STV voltages are fed rarely. This also may lead to BM depth instability and FOG bias.

Large setup time is caused by the heating of electronic components, their nonlinearity [5] and parasitic processes within the modulator of integrated optic chip (IOC). This paper discusses the methods to modify the existing data processing scheme of closed loop FOG.

OPEN LOOP FOG

A closed loop FOG containing fiber ring interferometer (FRI) and processing electronic module (EM) is described in [2, 3]. Figure 1 shows the open loop FOG, which also contains FRI and EM.

IOC is a multifunctional component, including Y-junction, polarizer and phase modulator.

Double-loop FOG is described in [2, 3]. The first loop (FB-1) compensates Sagnac phase difference, and the second loop (FB-2) compensates the temperature changes of IOC half-wave voltage. In this paper, a FOG without FB-1 loop is referred to as the open loop FOG.

Usually sinusoidal BM is implemented in open loop FOGs [4]. Standard rectangular BM in the form of voltage pulses with the frequency $1/(2\tau)$ (τ is the time of light running through fiber coil) is also possible [2, 3]. However, rectangular BM does not allow realizing FB-2 loop, so further we consider other type of rectangular BM. It is the so called low-frequency (LF) BM, which may have different patterns. Different forms of LF-modulation have been developed since 1990-ies [6, 7]. One of possible forms of the voltage for such kind of BM is shown in Fig. 2 (upper graph) [8]. Lower graph illustrates corresponding phase difference of the waves. Four values of BM depth $\pm(\pi \pm \Delta)$ rad are observed here. These are four operating points of FOG, corresponding to the same optical power level at photodetector. Here $\Delta = \pi/2^n$, with $n = 1, 2,$

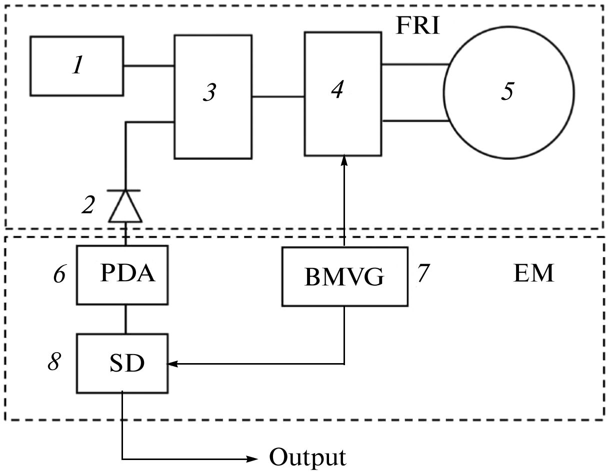


Fig. 1. Schematic diagram of open loop FOG, consisting of FRI and EM. *1*—optical power source, *2*—photodetector (PD), *3*—first fiber coupler, *4*—IOC, *5*—fiber sensing coil, *6*—photodetector amplifier (PDA), *7*—BM voltage generator (BMVG), *8*—synchronous detector (demodulator) of FOG rotation signal (SD).

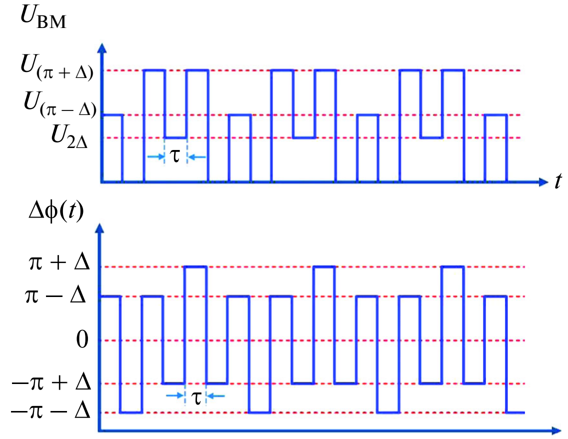


Fig. 2. Waveforms of BM voltage at the electrodes of IOC phase modulators (top) and of phase difference of FRI waves (bottom). U_x is the voltage corresponding to phase difference *x* rad.

3... The length of each BM voltage step is τ. BM frequency is $1/(6\tau)$ Hz, so this BM is called LF-modulation, as opposed to traditional modulation with the frequency $1/(2\tau)$ (HF-modulation). In the case of LF-modulation PDA output voltage goes to synchronous detector (SD):

$$U(t) \approx Q(t)\{1-\cos[\Delta(t)]\cos\Phi_S(t)\} \pm Q(t)\sin[\Delta(t)]\sin\Phi_S(t). \tag{1.1}$$

Here $Q(t) = P(t)\chi(t)G(t)$, $P(t)$ is the optical power of the waves interfering at photodetector, $\chi(t)$ is photodetector gain, $G(t)$ is PDA gain, Φ_S is Sagnac phase, and the sign of the second summand in the right-hand part is changed with the period of 3τ seconds. This summand with variable sign is FOG rotation signal. Procedure of rotation signal detection can be presented by the scheme (1) + (2) + (3) – (4) – (5) – (6), where (*k*) is the sample of the signal $U(t)$ at *k*-th τ-interval of one BM period ($k = 1...6$). Thus, it is clear that P, χ, G and Δ should be stable in time, which is possible only in warmed device under the stable environment.

Generally, the following can be written for rotation signal:

$$S_{rot}(t) = S_-(t) = \sum_{k=1}^{3} U(t+k\tau_1) - \sum_{k=4}^{6} U(t+k\tau_1) = 6Q(t)\sin\Delta(t)\sin\Phi_S(t), \tag{1.2}$$

where τ_1 is the time from the beginning of τ-interval, where signal $U(t)$ is sampled, to the sample (i.e. $\tau_1 < \tau$). Let parameters be changing in time: $Q(t) = Q_0 + \delta Q(t)$, $\Delta(t) = \Delta_0 + \delta\Delta(t)$, and $\delta Q(t) \ll Q_0$, $\delta\Delta(t) \ll \Delta_0$. To make the operational interferometric characteristic symmetrical relative to positive and negative rotation rates, BM with the depth $\Delta_0 = \pi/2$ rad should be applied, so

$$\delta\Phi_S(t) \approx \left[\delta Q(t)/Q_0 - \delta\Delta(t)^2/2\right]\tan\Phi_S(t).$$

Thus we have scale factor instability, which can be eliminated by stabilizing $Q(t)$ and $\Delta(t)$. It can be done using the dividing scheme [9], which utilizes an additional signal based on PDA output (1.1)

$$S_+(t) = \sum_{k=1}^{6} U(t+k\tau_1) \approx 6Q(t)\{1-\cos[\Delta(t)]\cos\Phi_S(t)\}.$$

Then $S_-(t)$ from (1.2) is divided by $S_+(t)$. Setting $\Delta(t) = \pi/2 + \delta\Delta(t)$ ($\delta\Delta(t) \ll \pi/2$), the following can be written:

$$\delta\Phi_S(t) = \delta\Delta(t)\sin\Phi_S(t).$$

It is seen that dividing scheme eliminates the effect of $Q(t)$ drift on the scale factor, but does not reduce the influence of BM depth variations $\delta\Delta(t)$. Moreover, dependence on $\delta\Delta(t)$ becomes linear. Therefore, BM depth should be stabilized separately using FB-2 loop, which is mentioned above. To realize this loop, a misalignment signal can be used. Figure 3 illustrates the generation of rotation signal (with stable modulator gain) and misalignment signal (with no rotation). Note that the ratio of misalignment signal amplitudes in open loop FOG is $A_+/A_- = 3$, as shown in Fig. 3. In closed FB-1 loop FOG described in section 2 any value of $\Delta = \pi/2^n$ rad ($n = 1, 2, 3,...$) can be used, for which the amplitude ratio of misalignment signal is

$$A_+/A_- = (\pi + \Delta)/(\pi - \Delta).$$

Rotation signal has the frequency $F_{rot} = 1/(6\tau)$ Hz and is detected according to the scheme (1) + (2) + (3) – (4) – (5) – (6). Misalignment signal has the frequency $F_m = 2F_{rot}$ and can be detected, for example,

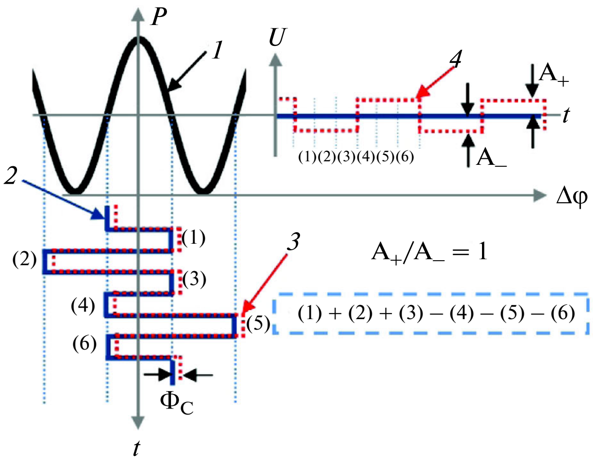


Fig. 3. Generation of FOG rotation signal (left) and misalignment signal (right). *1*—dependence of optical power *P* on phase difference Δφ of interfering waves, *2*—phase difference waveform (BM), *3*—change in Sagnac phase difference, *4*—change in voltage *U* at PDA output under changing Sagnac phase difference and BM (rotation signal), *5*—change in voltage *U* at PDA output under changing modulator gain and BM (misalignment signal), *6*—change in the phase modulator gain. Procedures for detecting the rotation and misalignment signals are given in dashed frames. Numbers in brackets are the numbers of six τ-intervals of BM period.

according to the scheme (2) – (1) + (5) – (4). Rotation and misalignment signals are described as follows:

$$S_{\text{rot}}(t) = S_-(t) = Q(t) + Q(t)\cos[\Delta\phi_M(t) + \Phi_S(t)],$$
$$U_{\text{mis}}(t) = Q(t) + Q(t)\cos\{[1+\varepsilon(t)]\Delta\phi_M(t)\},$$

where $\Delta\phi_M(t)$ is FRI waves phase difference due to BM voltage, $\varepsilon = \delta\eta/\eta_0$ is relative change in modulator gain. Modulator gain η is related to half-wave voltage V_π: $V_\pi\eta = \pi$. PDA output to be processed is given by

$$U(t) = Q(t) + Q(t)\cos\{[1+\varepsilon(t)]\Delta\phi_M(t) + \Phi_S(t)\} \neq S_{\text{rot}}(t) + U_{\text{mis}}(t). \quad (1.3)$$

This signal is not just a sum of rotation and misalignment signals. Its time dependence is shown in Fig. 4. From (1.3) with $\varepsilon \ll 1$,

$$U_{\text{mis}}(t) = S_{21} + S_{54} \approx 4\pi\varepsilon(t)Q(t)\sin\Delta_0\cos\Phi_S.$$

Here index 21 denotes the detection according to the scheme (2)–(1), index 54 is the same according to the scheme (5)–(4). This signal is zeroed by FB-2 loop, realized in EM digital part. It includes SD for misalignment signal (SD-2), regulator of BM voltage code and BM voltage generator.

FB-2 loop keeps the BM depth at the constant level at points $\pm\pi/2$ and $\pm3\pi/2$ rad. When BM depth is stable, rotation signal extracted from general signal (1.3) at PDA output is equal to (for $\varepsilon(t) \ll 1$)

$$S_-(t) \approx 6Q(t)\sin\Delta[1 - \varepsilon(t)(\pi/3 - \Delta)\cot\Delta]\sin\Phi_S.$$

For $\Delta = \pi/2$ rad, $S_-(t) \approx 6Q(t)\sin\Phi_S$, i.e. rotation signal is independent from modulator gain variations (this is also right for $\Delta = \pi/3$ rad, but does not provide scale factor symmetry for two signs of rotation rate).

Figures 5a, 5b present the experimentally obtained plots for open loop FOG with standard processing scheme and with dividing scheme. Optical power of Erbium Fiber Source (EFS) was changed within 30% relative to the mean value with frequency 0.022 Hz using the pump current. In a standard scheme (Fig. 5a) intensity variations lead to similar rotation rate variations, and dividing scheme (Fig. 5b) does not suffer from these variations. The difference between average rotation rates with and without dividing is caused by FOG axis misalignment during the experiments.

2. CLOSED LOOP FOG

For open loop FOG, Sagnac phase difference Φ_S is contained in all expressions as an argument of nonlinear trigonometric functions. To receive linear output over broad measurement range, compensating method was offered for reading Sagnac phase difference by closing the FB-1 loop [2, 3]. In this case PDA output can be written as

$$U(t) \approx Q(t)(1 - \cos\Delta) \pm \delta\varphi(t)Q(t)\sin\Delta \quad (2.1)$$

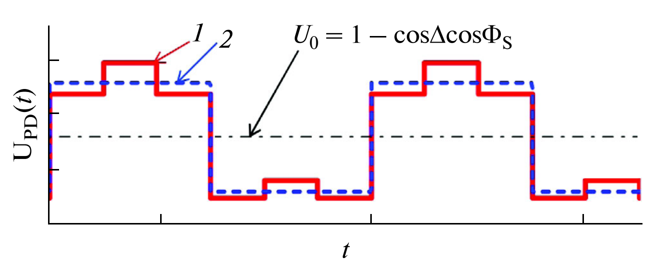


Fig. 4. PDA output with nonzero rotation rate and change in IOC modulator gain (*1*), rotation signal (*2*) is the signal (*4*) from Fig. 3. Signal (*1*) is the combination of signal (*2*) from this figure and signal (*5*) from Fig. 3.

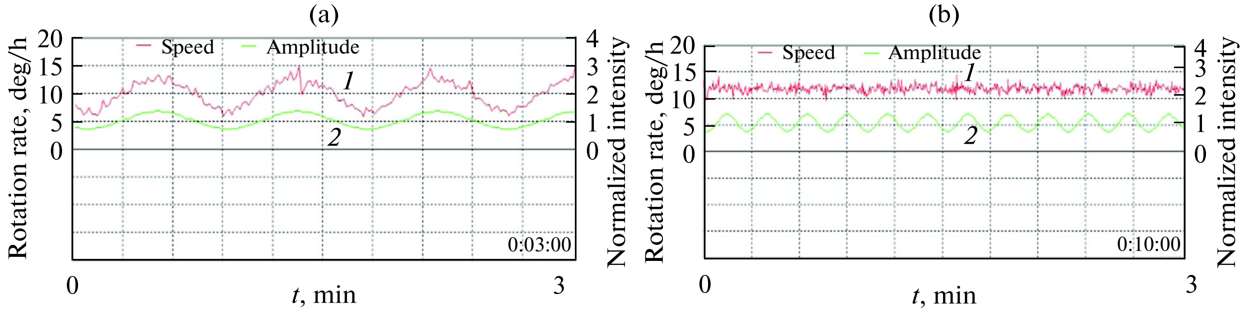


Fig. 5. (a) FOG output drift (*1*) with open first feedback loop (second feedback loop is closed, axis up): 3 min realization, signal dividing option is turned off; (b) (*2*) waveform of optical source output intensity normalized to unity: 10 min realization, signal dividing option is turned on.

where $\delta\varphi(t) = \Phi_S(t) - \Phi_K(t)$ is the error in Sagnac phase compensation, which replaces the Sagnac phase in all expressions for closed loop FOG. Here we assume that parameter Δ is stable in time due to FB-2 loop, operating according to the same scheme as for open loop FOG. For closing the FB-1 loop sawtooth voltage is applied to phase modulator simultaneously with BM voltage.

In closed FB-1 loop scheme [2], condition $\eta U_{res} = 2\pi$ is valid, the so called 2π-reset. Also two-phase BM scheme is known [3, 10], where during one phase (phase A) BM depth is $\pi/2$ rad, and during the other phase (phase B) it is $3\pi/2$ rad. As a result, the sum of BM and STV voltages introduces the phase difference not exceeding 2π rad (DAC overflow). This processing scheme will be further referred to as two-phase HF-modulation.

In the case of considered above low frequency BM, STV is also reset during DAC overflow, i.e. when the sum of BM and STV voltages exceeds $U_{2\pi}$. These moments correspond to BM voltage $U_{(\pi+\Delta)}$ (see Fig. 2). Then, instead of $U_{(\pi+\Delta)}$, $U_{(\pi-\Delta)}$ is applied to modulator (Fig. 2), after that BM is continued from this new value (BM time shift). Call this processing scheme LF-modulation with phase changes. However, in [5] it is shown that this scheme leads to appearance of dead zone, so STV is generated in a different manner for rotation rates about ± 1 deg/s (see section 2.2.).

2.1. SF Stabilization by Second FB Loop

It is known that in FOGs with closed FB-1 loop the scale factor is independent from variations of optical intensity, PD and PDA gains, at least if these variations are slow and small [9]. However, modulator half-wave voltage V_π strongly depends on temperature (~600–800 ppm/°C), therefore, scale factor will also be unstable under environmental temperature changes.

Data processing method for scale factor stabilization in [2] uses a rectangular pulse τ sec long, generated at the moment of STV reset when V_π is changed. In two-phase HF-modulation scheme when V_π is changing, a misalignment signal is generated with a period equal to STV period, when it has one sign during phase A and another sign during phase B of HF-modulation [10]. However, here the period of misalignment signal is determined by STV period, so at low rates scale factor instability may be significant.

As mentioned above, for LF-modulation in open loop FOG FB-2 loop can be realized when IOC modulator gain variations are monitored at each BM half-period independently of rotation rate. For closed FB-1 loop the same principle is possible for FB-2 loop [8]. In this case replacement $\Phi_S(t) \to \delta\varphi(t) \ll 1$ should be done in all above expressions, and misalignment signal will be written in the form

$$S_{mis}(t) \approx 4\pi\varepsilon(t)Q(t)\sin\Delta.$$

As a result, misalignment signal no more depends on rotation rate, unlike the open loop FOG. Thus closed FB-1 and FB-2 loops are almost independent from each other.

2.2. Dead Zone

Another problem of closed FB-1 loop FOG is the dead zone [3] appearing in two-phase HF-modulation scheme. Dead zone is caused by technological reasons such as imperfections of components and processing schemes, for example, electric cross-talk in SD-1 [11, 12]. Another possible source is parasitic Michelson interferometer due to back-reflections from joints of IOC waveguides with coil fiber [5, 6, 8].

According to the experiment, the described above LF-modulation with phase changes also suffers from technological dead zone. With scale factor of 20 μrad per 1 deg/s dead zone occurred about rotation rates of ± 0.7 deg/h. We think that this is due to the fact that here BM also has time-varying parameters (phase changes), similar to transitions between phases A and B in two-phase HF-modulation. Therefore, for low rotation rates BM with stable phase [13] is realized: BM voltage in Fig. 2 continues in time without changes.

Apart from the need to use BM with stable phase, DAC overflow should also be avoided, i.e., voltages applied to the modulator should not introduce phase difference over 2π rad. Two subranges: one for BM and one for STV should be introduced in range $U_{2\pi}$. Clearly, in this case amplitude U_{res} does not satisfy the condition $U_{res}\eta = 2\pi$ rad. However, STV amplitude in this scheme also does not lead to pulse of intensity at PD at STV reset. To find this condition, consider a more general expression for intensity P_{PD} at PD using the example of a simpler rectangular HF-modulation [2]. The difference between cosines of total phase difference at neighboring τ-intervals k and $k + 1$ can be written as

$$\cos[\delta\varphi + \theta_k \pm \Psi] - \cos[\delta\varphi + \theta_{k+1}]$$
$$= 2\sin\left[\frac{\theta_{k+1} - \theta_k \mp \Psi}{2}\right]\sin\left[\delta\varphi + \frac{\theta_{k+1} + \theta_k \pm \Psi}{2}\right] = 0,$$

where Ψ is the phase difference due to reset voltage U_{res}, θ_k and θ_{k+1} are phase differences due to HF-modulation at k-th and $(k + 1)$-th τ-intervals, and $\delta\varphi = \Phi_S - \Phi_K$ is the error of Sagnac phase compensation. Setting the first sinus in the right-hand part, which is independent from $\delta\varphi$, to zero, condition for STV reset is obtained:

$$\mp\Psi = 2\pi n + \theta_k - \theta_{k+1}.$$

Here the sign before Ψ is determined by the sign of rotation rate. Consideration of all reset conditions gives two STV reset conditions. The first one is

$$\Psi_1 = 2\Delta \tag{2.3}$$

for STV reset during the negative BM half-wave and at the positive rotation rate, or vice versa. The second condition is

$$\Psi_2 = 2\pi - 2\Delta \tag{2.4}$$

for STV reset during the negative BM half-wave and at negative rotation rate, or vice versa. For LF-modulation when processing the error signal at STV reset the sign of error signal should be changed during the time τ after reset. For minimal distortion of misalignment signal at STV reset, this information should be available when there are no distortions.

Currently, condition (2.3) is used. Maximum phase difference induced by the sum of BM and STV is

$$\Psi_1 + \pi + \Delta = \pi + 3\Delta,$$

which is always less than 2π rad, excluding $\Delta = \pi/2$ rad. Thus, for $\Delta < \pi/2$ rad part of the range 2π rad remains empty, allowing to avoid DAC overflow. In case of (2.4), maximum phase difference is

$$\Psi_2 + \pi + \Delta = 3\pi - \Delta,$$

which is always more than 2π rad. However, below it is shown that by changing the polarity of IOC modulator electrodes the required STV can be generated without DAC overflow. Note that now a so called overmodulation is used, i.e. BM with $\Delta = \pi/4$ rad (BM depth is $3\pi/4$ rad), which allows easy generation of STV according to (2.3). Then STV can be reset at any time independent from BM half-period.

According to (2.3), $\Delta = \pi/2$ rad is the only case when DAC overflow cannot be avoided. However, here triangular STV [13] can be realized (which should not be confused with dual-ramp voltage from [14]) shown in Fig. 6. Here, according to (2.3) a π rad reset occurs, and STV can be generated using the triangular STV with amplitude corresponding to phase difference $\pi/2$ rad. Here sum of BM and STV voltages will not lead to DAC overflow. By changing the electrode polarity, triangular STV with amplitude Δ can be transformed to a usual STV with amplitude 2Δ required according to (2.3).

For (2.4), vice versa, DAC overflow cannot be avoided except for the case $\Delta = \pi/2$ rad. However, it is possible with triangular STV and changing the electrode polarity. Here the sum of maximum BM and triangular STV voltages will equal the voltage inducing the phase difference $2(\pi - \Delta)/2 + (\pi + \Delta) = 2\pi$ rad.

For BM with $\Delta = \pi/4$ rad and stable phase, and for STV with reset voltage inducing the phase difference $\pi/2$ rad, dead zone was not observed, at least within ± 0.01 deg/h [5, 8]. This can be explained by the fact that detecting the electric cross-talk at PD by FB-1 loop gives only constant signal, which does not switch FB-1 loop into oscillation mode, thus, there are no reasons for the dead zone.

Note also that according to (2.3) STV amplitude is equal to difference between BM voltages giving phase differences $(\pi + \Delta)$ and $(\pi - \Delta)$ rad, so in practice it's better to set the value U_{res} by the difference of corresponding BM voltages $U_{(\pi + \Delta)} - U_{(\pi - \Delta)}$.

However, from (2.3), STV amplitude is lower than standard value 2π rad, which reduces the measurement range of rotation rates. Then it is reasonable to divide this range into two subranges. In the first, BM with stable phase should be used, thus excluding the dead zone. In the second subrange, BM with phase changing should be used. At present, Sagnac phase difference crossing π rad is counted, so the dynamic range is over ± 1000 deg/s for coil with fiber length 500 m and average diameter ~500 mm. This principle can be used to extend FOG dynamic range up to the limits set by the source spectral width [15].

2.3. Setup Time

One more problem encountered in FOGs is the large setup time and signal drift under temperature fluctuations or other external influences. This drift may be caused by polarization effects and time-varying thermal fields (Shupe effect). These effects are not considered here.

Variations of optical source power, or, more generally, variations of FB-1 loop gain are manifested by smoothing of rectangular BM fronts due to finite capacity and resistance of modulator metal electrodes

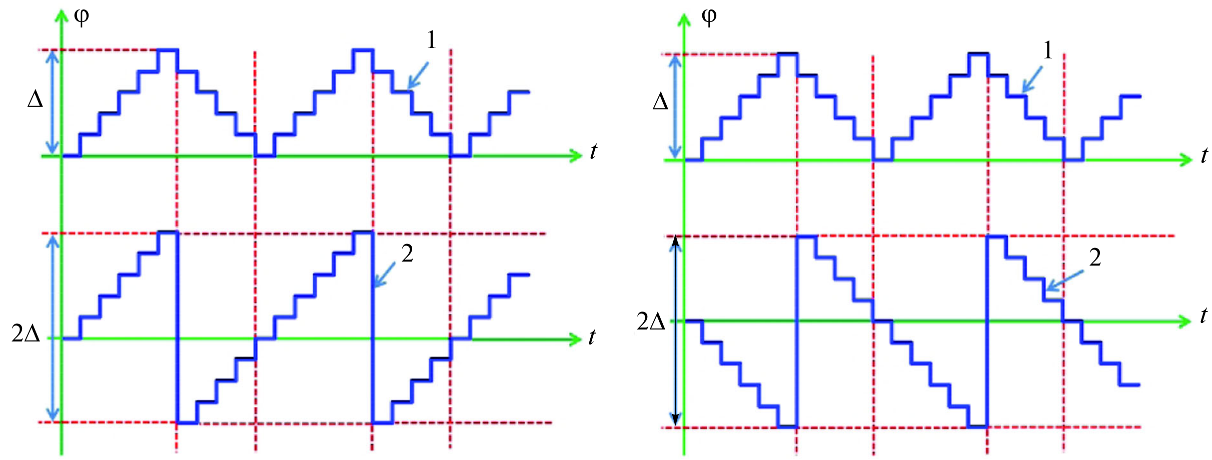


Fig. 6. Transformation of triangular STV 1 into usual STV 2 with only rising (left bottom) or falling fronts (right bottom) by varying the polarity of electrodes of phase modulators. Vertical dashed lines illustrate the moments of changing the electrodes polarity.

(parasitic HF-dynamics of modulator) [16]. Front smoothing means that after BM voltage jump the phase difference reaches the required value only after a certain time. This smoothing leads to peaks in PDA output (Fig. 16). In [17] it is offered to remove these peaks because further electronic components widen them strongly. However, if the peak length is small compared to τ, they are removed according to [17], after that signal samples are taken closer to the end of τ-intervals. The experiments still reveal that even after this measure initial long-time drift is large because it is based on a different low-frequency process. It also distorts BM voltage pulses throughout the whole τ-interval. These distortions are due to dependence of a half-wave modulator voltage V_π on the frequency of applied voltage [18, 19]. This dependency comes from charge trapping in Lithium Niobate, which is influenced by humidity, temperature, pressure, radiation [20] and IOC aging. In [18, 19] these processes are described by the transfer function of the form

$$h(s) = (s + a)/(s + b),$$

where a and b are zero and pole of IOC transfer function. Typical graph of simulated phase difference of FRI waves is shown in Fig. 7, top.

Figure 7, bottom shows cosine vs distorted phase difference. According to this model, distorted PDA output is the same over both BM half-periods, so during the detection of error (rotation) signal these distortions do not introduce the errors into FOG output. Nevertheless, experiment reveals significant initial drift of FOG output (setup time), along with constant bias. Initial drift can be explained by the fact that during FOG heating IOC parameters change, so the distorted PDA output becomes asymmetric over two BM half-periods, giving the parasitic signal which leads to rotation rate error.

As for misalignment signal, it has non-zero parasitic value, because signals at first and second, and at fourth and fifth τ-intervals in sampling points *7* and *8* are different. Clearly, information on distorted PDA output is lost except for points *7* and *8*, so FB-2 loop interprets this signal as some misalignment signal *4* and sets it equal to zero. Here stabilization occurs at a shifted level 6 instead of initial operating level *5*. This may also serve an explanation of FOG bias, and when PDA output distortions vary this may explain the initial FOG output drift.

This explanation of initial FOG drift suggests that it is correlated with variations of IOC voltage code [20], which is confirmed by numerous experiments. Consequently, initial drift can be eliminated during data processing by correcting FOG output by IOC voltage code, which also drifts while being regulated by FB-2 loop, similarly to initial FOG output drift.

Figure 8 shows the example of such correction, where *1* is the initial FOG output, and *2* is the output corrected by the voltage code at IOC modulator (coil scale factor is 0.0052 rad per 1 deg/s).

Figure 8, right shows PDA output with open FB-2 loop, revealing rather bad state of IOC from the viewpoint of parasitic LF-processes (top graph is simulated PDA output, bottom graph is the measured one). Signal *4* is the modification of signal *3* from Fig. 7 when FB-2 loop is closed. The plots show that FOG output can be corrected even with unsatisfactory IOC. FOG output is recorded and correcting coefficient Q is calculated to correct zero drift:

$$\delta\Omega(t) = Q\left[\delta U(t) - \delta U_\eta\right].$$

Here $\delta U(t)$ describes the changes of IOC voltage code due to LF-process and temperature dependence of V_π, δU_η is conditioned only by temperature dependence of V_π, so with $T = \text{const}$ $\delta U_\eta = \text{const}$. In the future double correction can be provided with account for the temperature. When the temperature changes then, firstly, δU_η changes, and secondly, gain η_0 changes, which

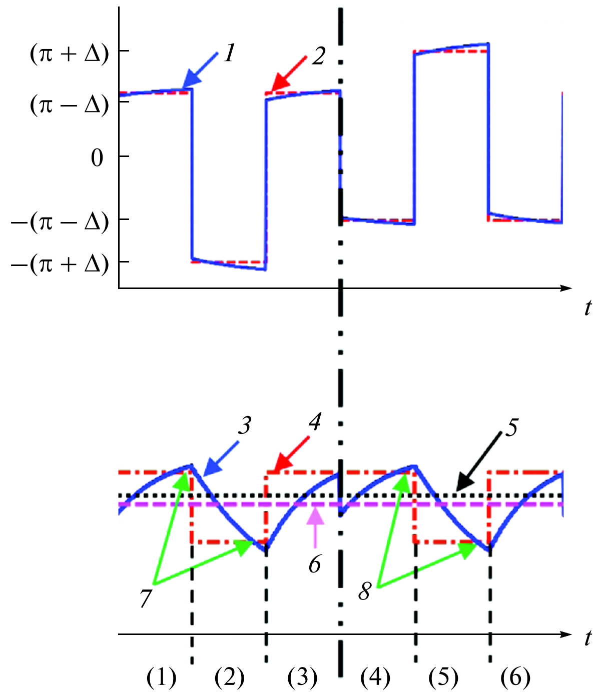


Fig. 7. Top: *1*—phase difference of FRI waves distorted by modulator parasitic LH-dynamics, *2*—ideal phase difference. Bottom: *3*—PDA output with BM voltage distorted by modulators, *4*—effective misalignment signal, corresponding to distorted PDA output, *5*—FOG operating level with no parasitic LF-dynamics in modulators, *6*—new operating level, where FOG is put by FB-2 loop under parasitic LF-dynamics in modulators, *7*— sampling points of misalignment signal at the first and second τ-intervals, *8*—sampling points of misalignment signal at the fourth and fifth τ-intervals. Numbers in brackets are the numbers of six τ-intervals of one BM period.

can be described by linear approximations (T_c is the room temperature):

$$\eta_0(T) \approx \eta_0(T_c) + \beta(T - T_c), \quad \delta U_\eta(T) \approx U_0\beta(T - T_c).$$

As a result, the following can be written for FOG output error:

$$\delta\Omega(t) \approx Q[1 + \beta(T - T_c)/\eta_0(T_c)] \times \{\delta U(t) - U_0\beta[T(t) - T_c]\}.$$

Thus one more coefficient β appears for initial drift correction (double correction). To realize this correction, the temperature should be measured at each moment of time.

Figure 9 shows Allan variances for plots 1 and 2 from Fig. 8.

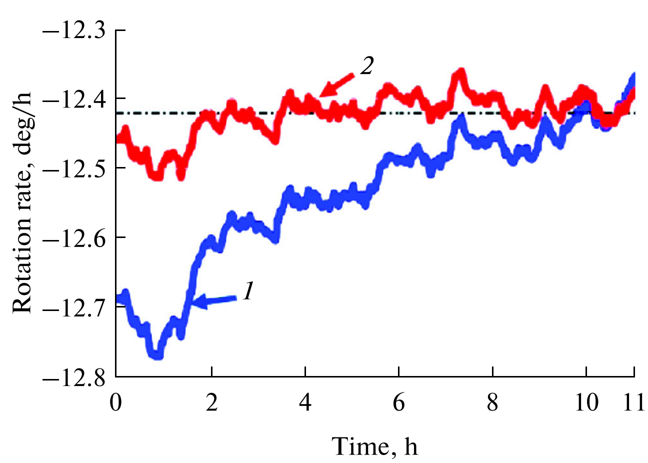
Bias stability in graph *1* is 0.02 deg/h, linear drift is 0.05 deg/h/h. For graph *2* the bias drift is at least ten times less. Linear drift is characterized by Allan variance bending up at final section with +1 slope. Graph 2 has lower slope in final section, which suggests that it is not a linear drift, but the beginning of some quasiperiodic process, which can be identified with longer realization. Probably it is due to the fact that the signal is undercorrected in some time sections and overcorrected in the other, because FOG output cannot behave exactly as IOC voltage.

Figure 10 presents the graphs similar to those in Fig. 8 but for a different IOC. It is seen that, first, initial drift here is much lower compared to Fig. 8; second, correction with IOC voltage code is also possible and improves the result (though not so strongly); third, condition of IOC modulators is much better than in Fig. 8, which is the reason of improvement. This example can also be considered an experimental illustration of performance of processing scheme with constant phase BM described in part 2.

Note that described correction is also applicable to open loop FOGs.

PERSPECTIVES

The considered method of correcting the initial drift by IOC voltage code is the simplest one; it does



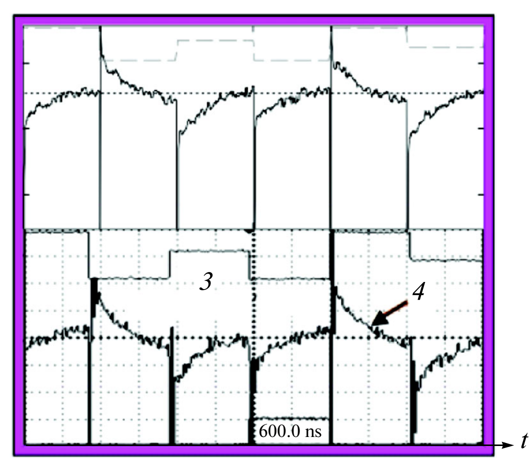


Fig. 8. Initial drift of rotation rate (*1*), rotation rate corrected by IOC voltage code (*2*). Right: *3*—BM voltage, *4*—PDA output.

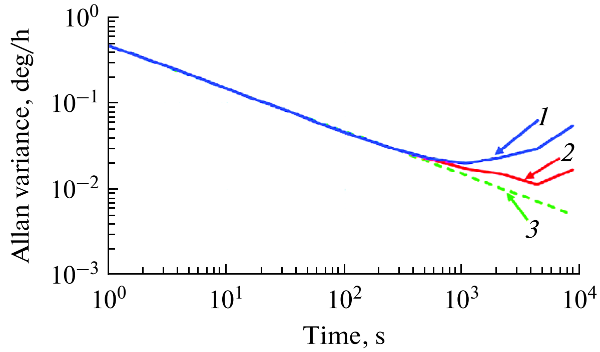


Fig. 9. Allan variances for non-corrected (*1*) and corrected (*2*) rotation rate (plots *1* and *2* in Fig. 12). Line *3* is the white noise.

not require any component or program additions in both FB loops, or introduction of additional FB loops. Further improvements will require such additions.

Above, initial drift is explained by the shift of FOG operating level when IOC parasitic LF-process influences the operation of FB-2 loop. This shift is absent only if the effective misalignment signal *4* in Fig. 8 has the ratio of pulse amplitudes relative to operating level equal to $(\pi - \Delta)/(\pi + \Delta)$ (see Fig. 3). Figure 11 shows the simulated PDA outputs during one half-period of BM for coils with fiber length 5000 m and 500 m. In the figure, misalignment signals *2* are associated with distorted signals with the required amplitude ratio A_+/A_-. At the sampling points in the first case the amplitude ratio of signal *1* at PDA satisfies this condition, unlike the second case. Thus, operating level is not shifted in the first case, and it is in the second. When the parameters of distorted PDA output are slightly changed the situation in the first case remains the same (no drift), and in the second case it changes, which means the floating of operating level in time. In the context of the offered explanation for initial drift, this floating is the source of this drift.

Unfortunately, the second case occurs more often. Here, with standard operation of FB-2 loop, when all voltages are multiplied by the same number, it is impossible to maintain the device at necessary operating level. This can be done by changing the ratio of BM voltages at different τ-intervals in order to overcome the limitation of the ratio $A_+/A_- = (\pi - \Delta)/(\pi + \Delta)$. Figure 12 shows the arrangement of FOG operating levels (2 and 3) on cosine curve with complete and incomplete compensation of parasitic misalignment signal.

Figure 13 shows the following graphs: initial BM voltage (1); BM voltage with changed amplitude ratio of pulses with length τ (2); change of voltage codes, which compensates the parasitic misalignment signal (3); initial phase difference of FRI waves (4); phase difference (5) corrected by the voltage *3*.

Waveform of BM voltage codes for compensation of parasitic misalignment signal (voltage 3) is a separate STV with rising and falling fronts. Voltage 3 should not be confused with triangular STV in Fig. 6, because it does not depend on rotation rate and has a fixed period of 6τ s. Voltage step 3 has the same value at all τ-intervals. To change the code ratio, amplitude of voltage *3* is changed. It leads to changes in BM voltage (dashed line *2*) and phase difference of FRI waves (dashed line *5*). As seen from Fig. 13, BM depth $\pm(\pi - \Delta)$ is decreased, and BM depth $\pm(\pi + \Delta)$ is increased by the same amount Δ_0. When Δ_0 changes its sign, amplitude of voltage 3 changes its sign, too. Thus, zero Δ_0 can be reached by varying the amplitude of voltage 3 using the ratio of BM voltage codes. This would keep the device at the required operating level. In this case the amplitude of STV, which compensates Sagnac phase, is determined as the difference of BM voltages $(\pi + \Delta)-(\pi - \Delta) = 2\Delta$rad, according to (2.3).

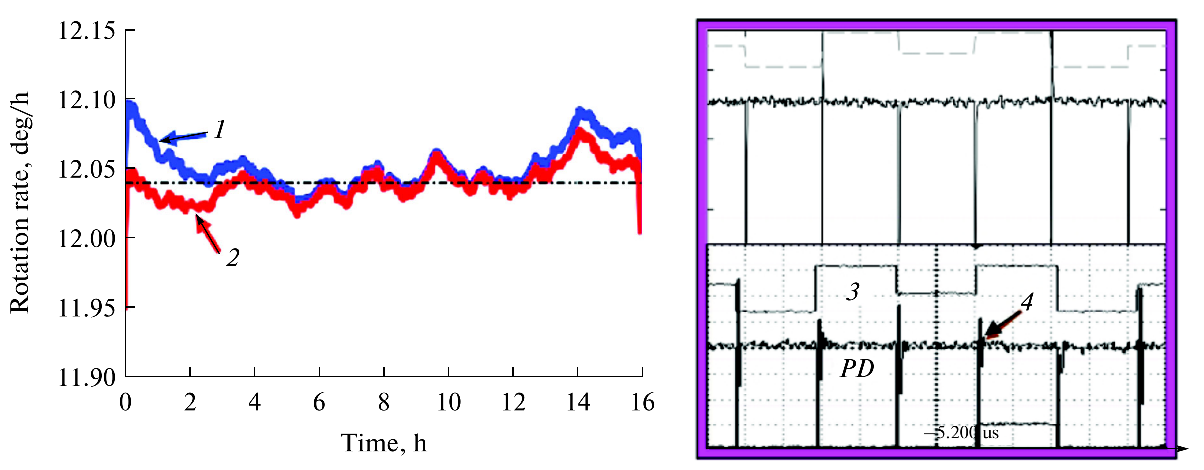


Fig. 10. FOG output drift (*1*), FOG output corrected by IOC voltage code (*2*). Right: (*3*) is BM voltage, (*4*) is PDA output.

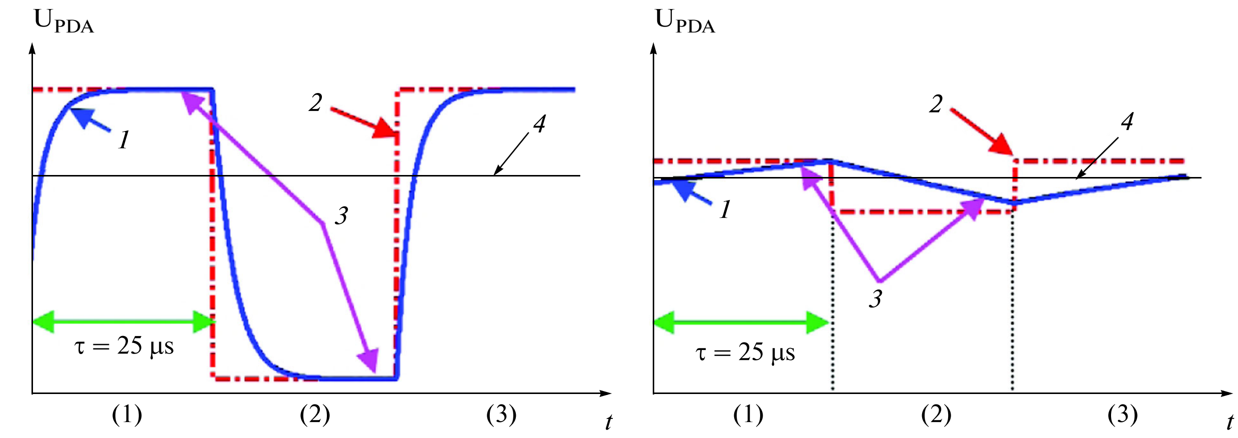


Fig. 11. PDA output for fiber coils 5000 m and 500 m long. *1*—PDA output, *2*—effective misalignment signal corresponding to PDA outputs, *3*—sampling points in 1st and 2nd τ-intervals, *4*—FOG operating point for perfect modulator. Numbers in brackets correspond to τ-intervals.

In practice BM voltage codes can be varied using the third FB loop, which monitors the resets of STV compensating the Sagnac phase. Figure 14 shows graphs of FOG output drift during 1 sec measurement of Earth rotation rate with initial BM code ratio $\pm(\pi \pm \Delta)$ (graph A) and with changed ratio (graph B). Peaks 1 in the graphs correspond to STV resets. Clearly, peaks 1 are significantly reduced by changing of BM codes. Code of STV voltage at $\Delta = \pi/4$ rad (see 2.3) in our experiment was set in the form of $K_0/4$, where K_0 is the maximum DAC code corresponding to $U_{2\pi}$ when FB-2 loop is closed.

Value $K_0/4$ thus should correspond to STV amplitude, introducing the phase difference $\pi/2$ rad. However, the peaks in signal in graph A (Fig. 14) mean that $\Delta \neq \pi/4$ rad (condition (2.3) is not observed). This reveals the unsatisfactory state of modulators (parasitic effects), confirmed by PDA output (Fig. 15).

Figure 16 shows the results of measuring the Earth rotation rate using IOC modulators in satisfactory state (small parasitic effects). This is revealed by the absence of peaks in FOG output: here $\Delta = \pi/4$ rad, so FOG operating points are at the required operating level (line *5* at Fig. 8) with the working FB-2. LF-process in IOC modulator can be simulated by changing the BM voltage codes. Then voltage peaks are observed in FOG output, which are generated due to reset of STV compensating Sagnac phase.

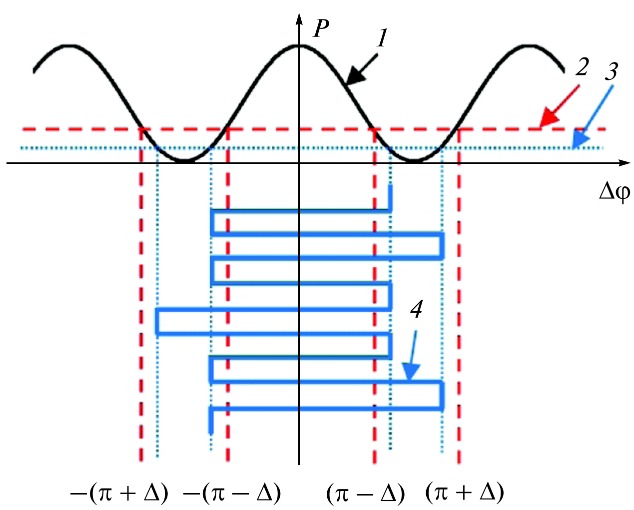


Fig. 12. *1*—interfering cosine curve, *2*—position of operating points for complete compensation of parasitic misalignment signal, *3*—position of operating points for incomplete compensation by second FB loop, *4*—phase difference of ring interferometer waves.

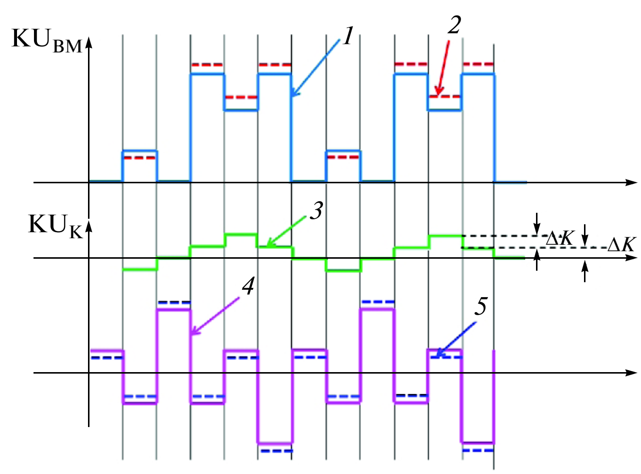


Fig. 13. Waveform of BM voltage codes for compensation of parasitic misalignment signal.

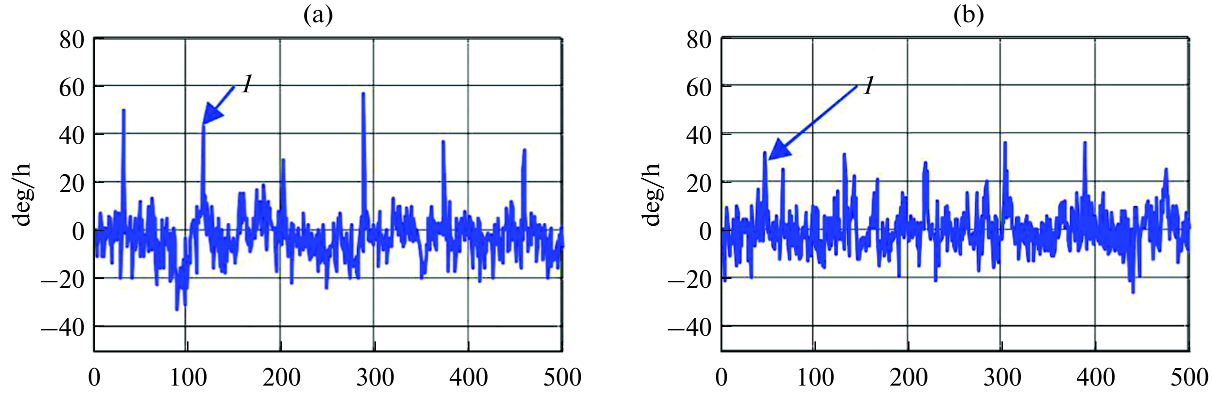


Fig. 14. FOG output drift during 1 s for initial (a) and changed (b) ratio of BM codes, *1*—peaks during STV reset.

Therefore, the described scheme is intended to improve the FOG accuracy (enhance the stability of scale factor and rotation signal), suffering from parasitic processes in IOC layer. This is a merely electronic correction method, being an alternative for previously described variants with correction using several temperature sensors in different points of FOG housing [21].

Finally note that parasitic processes in the modulator can make FOG output sensitive to optical source intensity variations (for example, during its initial self-heating). In this case detected rotation rate is related to its true value as

$$\Omega_{meas}(t) = \Omega_{rot}(t) + \theta(t)/[2Q(t)\sin\Delta],$$

where $\theta(t)$ is the value determined by parasitic processes in modulators. Value $\theta(t)$ can be treated as parasitic bias at SD-1 output, which detects the error signal (of rotation signal in open loop FOG). According to (2.1), value $Q(t)$, containing the intensity variations, can be extracted by another SD in the form

$$S_+(t) = U(t + \tau) + U(t) \approx 2Q(t)(1 - \cos\Delta).$$

This signal can be used to construct the fourth FB loop to stabilize $Q(t)$, for example, in the scheme with parallel ADC by varying the number of samples, or by using the dividing scheme eliminating the dependence on $Q(t)$ described in Section 1.

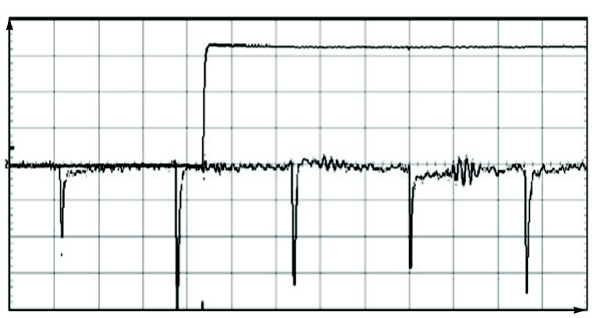


Fig. 15. PDA output waveform.

CONCLUSIONS

The paper describes the concept of digital open loop FOG with low-frequency rectangular phase modulation and improved scale factor stability. The latter is reached using stabilization of the rotation signal amplitude by dividing it by the constant component of optical intensity at photodetector. Stability of biasing modulation depth is reached by realizing the feedback loop. Initial parasitic bias of FOG output may be corrected using parasitic bias at the output of synchronous detector of misalignment signal. The concept of FOG with two feedback loops is described (for compensating Sagnac phase with sawtooth voltage and for stabilizing biasing modulation depth and scale factor). In dual-loop FOG, the dead zone is eliminated and setup time is reduced. The dead zone is eliminated using stable parameters of biasing modulation voltage in time and by using sawtooth voltage with amplitude depending on modulation depth. Scale factor stability at low rotation rates is improved due to the fact that the second feedback loop works not with sawtooth voltage resets, but with high-frequency misalignment signal.

A method to correct FOG output during setup is proposed. Setup is conditioned by the influence of parasitic processes in integrated optic modulator. Voltage code at IOC phase modulator serves as a correcting signal. As a perspective for further improvements of FOG accuracy, a method to improve the stability of scale factor and rotation signal amplitude is presented. The method is based on compensation of parasitic effects in IOC phase modulators. Parasitic effects are suppressed using the third feedback loop, which keeps the signal stable during resets of sawtooth voltage compensating the Sagnac phase by changing the ratio of biasing modulation amplitudes $(\pi - \Delta)/(\pi + \Delta)$.

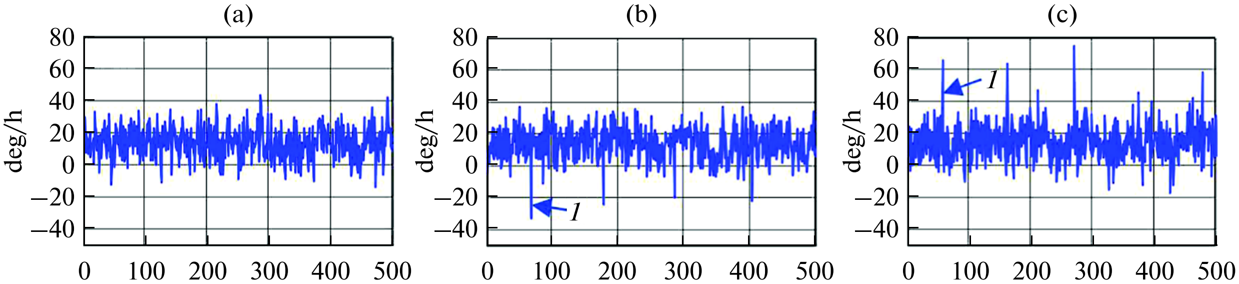


Fig. 16. Measurements of Earth rotation rate. (a)—conditions (2.3)–(2.4) for reset of STV compensating Sagnac phase are met, (b) and (c)—BM code ratio is changed, conditions are not met, *1*— peaks caused by STV reset in these cases.

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